Available online at www.sciencedirect.com
ScienceDirect
Electrochimica Acta 53 (2007) 990–997
ELECTROCHIMICA Acta
www.elsevier.com/locate/electacta

A DFT study of electronic structures, energies, and molecular properties of lithium bis[croconato]borate and its derivatives

Zhao-Ming Xue [a,b,*], Yu-Zhen Ding [a], Chun-Hua Chen [b]

[a] Department of Chemistry, Anhui University, Hefei, Anhui 230039, PR China
[b] Department of Materials Science and Engineering, University of Science and Technology of China, Hefei, Anhui 230026, PR China

Received 3 May 2007; received in revised form 8 August 2007; accepted 8 August 2007
Available online 14 August 2007

Abstract

Theoretical studies on electrolyte salts, lithium bis[croconato(2-)]borate (LBCB) and its derivatives, lithium [croconato(2-) salicylato(2-)]borate (LCSB), and bis[salicylato(2-)]borate (LBSB) are carried out using density functional theory (DFT) method and B3LYP theory level for the first time. Bidentate structures involving two oxygen atoms are preferred. Based on these conformations, a linear correlation was observed between the highest occupied molecular orbital (HOMO) energies and the limiting oxidation potentials measured by linear sweep voltammetry, which supports experimental results that strongly electron-withdrawing substituent anions are more resistant against oxidation than their organic counterparts. The correlations were also observed between ionic conductivity and binding energy, solubility and theoretical set of parameters of anion, thermal stability and the hardness (η). Wave function analyses have been performed by natural bond orbital (NBO) method to further investigate the cation–anion interactions.

Keywords: DFT; Lithium salts; Borate; Electronic structures; Energies; Molecular properties

1. Introduction

Many researchers have attempted to develop improved electrolytes for lithium batteries with high energy density and good rechargeability [1–4] as power sources for portable electronic instruments and electric vehicles in order to protect the environment and natural resources. Barthel et al. reported a new class of electrochemically and thermally stable lithium salts [5,6] with a chelate-type boron-containing anion, such as lithium bis[1,2-benzenediolato(2-)-O,O′]borate (LBBB) [7], lithium bis[2,3-naphtha-lene-diolato(2-)-O,O′]borate (LBNB) [8,9], lithium bis[2,2-biphenyldiolato(2-)-O,O′]borate (LBBPB) [10], and lithium bis[salicylato(2-)]borate (LBSB) [10]. Recently, Xu and Angell reported an advanced electrolyte of Li-ion battery, lithium bis(oxalate)borate (LiBOB) [11]. The common feature of these anions is an extensive charge delocalization in their anions caused by strongly electron-withdrawing substituents. Due to this particular feature, the lithium salts of these anions yield sufficiently high ionic conductivity solutions, and exhibit wide electrochemical stability windows and high thermal stability.

The focus on the development of the novel lithium salts has been placed on seeking proper anions for coordination with lithium cation to obtain desired species with appropriate chemical and electrochemical property. This is usually time-consuming and sometimes even clueless. To save both time and resources and to provide a guideline to experimental studies, a theoretical treatment on predicting the structure and (electro) chemical characteristics is undoubtedly necessary and important. Because the anion–cation interaction within the lithium salts plays an important role in determining to the solubility, ionic conductivity, electrochemical windows and thermal stability [12–14], a thorough computational investigation on the electronic structures, energies, and orbitals of this B-containing lithium salts would be desirable to better our understanding their properties at a quantum chemistry level.

According to our previous theoretical researches [15–19], the strongly electron-withdrawing anion, $C_5O_5^{2-}$[dianion of croconic acid (4,5-dihydroxycyclopent-4-ene-1,2,3-trione)], was

* Corresponding author at: Department of Chemistry, Anhui University, Hefei 230039, PR China. Fax: +86 551 5106000.
E-mail address: zmxue@ahu.edu.cn (Z.-M. Xue).

0013-4686/$ – see front matter © 2007 Elsevier Ltd. All rights reserved.
doi:10.1016/j.electacta.2007.08.013

chosen as the chelator to coordinate with boron to form new lithium salts, lithium bis[croconato]borate (LBCB) and its novel derivative, lithium [croconato salicylato]borate (LCSB), which yield high ionic conductivity solutions, and exhibit wide electrochemical stability windows and high thermal stability [20].

In this study, to further our understanding the properties of these new lithium salts at a quantum chemistry level, the density functional theory (DFT) B3LYP method [21,22] was chosen because it accounts better for electron correlation energies and greatly reduces the calculation expenses with rather reliable description of both geometries and energy of targeted molecules. We will address the following issues regarding the BCB^-, CSB^- and BSB^- anions and their lithium salts: (a) the conformational characteristics of the three anions, (b) molecular structures and the electronic distributions of the stable Li^+BCB^-, Li^+CSB^- and Li^+BSB^- ions pairs, (c) the nature of the bonding corresponding to each local energy minimum by estimating electrostatic and charge transfer contributions to the binding energy, and (d) the relationship between molecular structures and properties.

2. Calculation and experimental methods

The local minima of the sextuple complexes have been fully optimized by analytic gradient techniques. The method used was the density functional theory (DFT) with Becke's three parameters (B3) exchange functional along with the Lee–Yang–Parr (LYP) non-local correlation functional (B3LYP). All of the complexes were treated with DFT method at B3LYP/6-31+G(d,p) level for full geometry optimization. Nature population analysis (NPA) and natural bond orbital (NBO) [23] analysis were performed at the same level using NBO program to obtain quantitative analysis of cation–anion interactions between NBOs of the complexes.

Single-point energy calculations at higher level basis set (B3LYP/6-311++G (3df, 3pd)//B3LYP/6-31+G(d,p)) were also performed to obtain more accurate binding energies, which were defined as $E_{\text{bind}} = E_{\text{metalcomplex}} - (E_{\text{cation-free}} + E_{\text{anion-free}})$. The HF method at the same level computation was performed for comparing with DFT at some cases. All of the HF, DFT and NBO calculations were performed using the Gaussian 03 program package [24].

The purification procedures for propylene carbonate (PC), ethylene carbonate (EC), 1,2-dimethoxyethane (DME), acetonitrile (AN), and tetrahydrofuran (THF), as well as the electrochemical equipment for electrochemical studies, are given in Refs. [7,8].

Thermogravimetric (TG) analysis of the lithium organoborates was carried out with Perkin-Elmer Pyres-1 DMDA-V 1 model, using a sample of about 10 mg. The NMR spectra were measured with DMX-500. Inductively coupled plasma (ICP) emission spectrometry for both Li and B (model Poasma—Spec). The decomposition voltages (i–E curves) of the electrolytes using a three-electrode system (stainless steel plate working, Li plate counter, and Li plate reference electrodes)

Table 1
Optimized geometries of the free anions, their lithium ion pairs[a,b]

	Anion			Ion pair		
	BSB^-	CSB^-	BCB^-	LBSB	LCSB	LBCB
r(B1–O2)		1.533	1.498		1.544	1.566
r(B1–O3)		1.530	1.498		1.636	1.524
r(B1–O4)	1.483	1.445		1.520	1.409	
r(B1–O5)	1.463	1.443		1.433	1.406	
r(O2–C18)		1.314	1.323		1.293	1.282
r(O3–C14)		1.314	1.323		1.271	1.301
r(O4–C7)	1.330	1.341		1.323	1.365	
r(O5–C8)	1.336	1.342		1.345	1.361	
r(O6–C7)	1.226	1.220		1.250	1.211	
r(C8–C9)	1.409	1.406		1.403	1.399	
r(C8–C13)	1.412	1.409		1.416	1.406	
r(C14–C15)		1.457	1.458		1.484	1.427
r(C14–C18)		1.390	1.384		1.425	1.416
r($Li–O_a$)				1.873	1.867	1.880
r($Li–O_b$)				1.873	1.909	1.926
α(O_a–Li–O_b)				140.7	94.0	92.9
α(O2–B1–O3)		102.6	104.6		108.3	101.4
α(O2–B1–O4)		109.6			110.2	111.0
α(O4–B1–O5)	112.5	115.3		111.3	119.2	107.8
d(O3–B1–O2–C18)		0.2	0.0		1.5	0.0
d(O5–B1–O4–C7)	24.5	2.1		50.5	10.2	
d(O2– B1–O4–C7)		121.9			139.4	
d(C8–C9–C10–C11)	0.2	0.0		0.6	0.0	
dC15–C14–C18–C17)		0.0	0.0		0.0	0.0

[a] Bond lengths in Å, bond angles and dihedral angles in degrees.
[b] Optimized with the DFT method at B3LYP/6-31++G(2df,2p) level.

were measured at a scan rate of $9\,\mathrm{mV\,s^{-1}}$. Preparation of the electrolyte solutions and the cell assembly were carried out in a glove box (Labmaster 130, MBRAUN).

3. Results and discussion

3.1. Geometries

First, we shall present and discuss the results obtained for the "model" system CSB^- and Li^+CSB^-. The other two systems studied here will be discussed with reference to this first system. The lithium bis(trifluoromethansulfonyl) imide (LiTFSI) ion pair was studied by Arnaud et al. [25] using *ab initio* Hartree-Fock method up to 6-31+G* level. Their work showed that the bidentate structures are the most stable. Francisco and Williams [26] and Spoliti et al. [27] reported that the total energy order for lithium tetrafluoroborate ion pairs is bidentate < tridentate < monodentate, and a similar energy order was reported for the lithium perchlorate ion pairs [28]. Apparently, a common feature exists for the energy of these lithium complexes, i.e. the bidentate structure has the lowest total energy calculated for these isolated ion pairs. Thus, only the most stable optimized geometries of the bidentate ion pairs (one of the three conformers of the LBSB, one of the four conformers of the LCSB, one of the three conformers of the LBCB) are given in Table 1 and Fig. 1.

The CSB^- belongs to the $C_1[O(B),2SGD(C_6H_4O_2)]$ point group; the equilibrium B–O2, B–O3, B–O4, B–O5 distances are 1.533, 1.530, 1.445, and 1.443 Å, respectively. The O2–C18, O4–C7 is 1.314 and 1.341 Å. The CSB^- clearly has no asymmetrical structure. Each of the moieties ($C_7H_4O_3$, C_5O_5) has a planar structure, and the two planes are perpendicular to each other. Compared with free CSB^-, structural changes of the CSB^- moiety in the divalent salt are as follows: (1) as expected, the B1–O4(5) in cation-unbounded moiety($C_7H_4O_3$) distances are shortened by 0.036 Å, whereas the B1–O2(3) in cation-bounded moiety (C_5O_5) distances are lengthened by ca. 0.05 Å. The BO(2,3,4,5)–C(7,8,14,18) distances are shortened by ca. 0.02 Å, the C14–C(15,18) distances are lengthened by ca. 0.03 Å; (2) the O2–B1–O3 and O4–B1–O5 angles increase by ca. 5° indicating that a charge delocalization occurs when CSB^- coordinates with lithium cation; (3) except for the phenylic and croconic rings keeping in planar surfaces, the geometry of the CSB^- is severely distorted for the LCSB ion pair shown in Fig. 1 because of the strong interaction between the lithium cation and the BBB^-. The O2–B1–O4–C7 dihedral angle increases by 18°, the O5–B1–O4–C7 dihedral angle increases by 8°; (4) the equilibrium Li–O20(21) distance is longer than that obtained for $BF4^-\cdots Li^+$ which lies in the range 1.750–1.813 Å according to the level of calculation [26].

Compared with free CSB^-, structural changes of the BSB^- and BCB^- are as follows: (1) B–O distances are shortened by ca. 0.032–0.035 Å, and the O–C (14 or 18) distance is lengthened by ca. 0.01 Å in BCB^-, whereas the B–O distances is lengthened by ca. 0.02–0.04 Å, and the O–C (7 or 8) distance is shortened by ca. 0.01 Å in BSB^-; (2) with the increase of the number of C_5O_5, in the order BSB^-, CSB^-, BCB^-, the C14–C18 and C8–C9 decrease by ca. 0.03–0.06 Å. the C7–O6 distance in CSB^- is shorter than that in BSB^- by 0.006 Å.

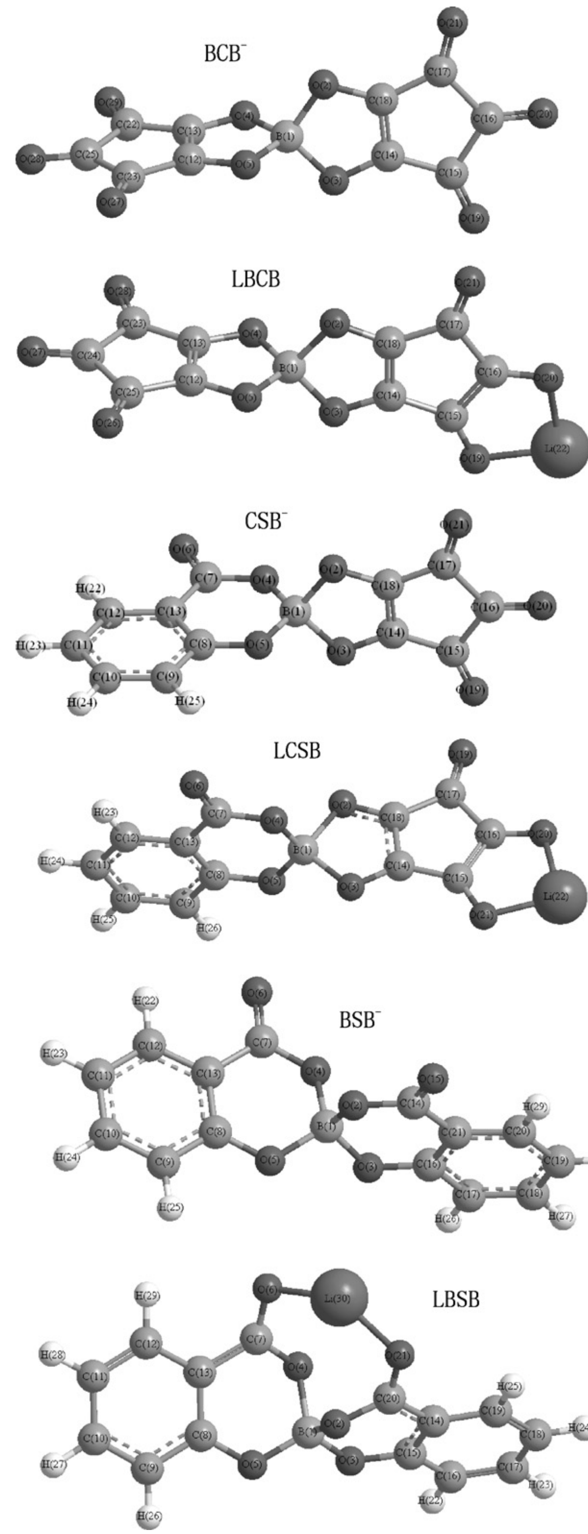

Fig. 1. Optimized geometry of BCB^-, CSB^-, BSB^-, LBCB, LCSB and LBSB.

Compared with LCSB, structural changes of the LBSB and LBCB are as follows: (1) with the increase of the number of C_5O_5, in the order BSB^-, CSB^-, BCB^-, B–O distances are shortened, but the BO–C distances is lengthened; (2) with the

Table 2
NPA atomic charges of the free anions, their lithium ion pairs

	BSB^-	CSB^-	BCB^-	LBSB	LCSB	LBCB
Li				0.9372	0.9502	0.9525
B	1.3069	1.3018	1.2984	1.3072	1.3108	1.2993
O2		−0.6367	−0.6498		−0.5921	−0.5617
O3		−0.6366	0.8498		−0.5468	−0.6011
O4	−0.7174	−0.7196		−0.7199	−0.7381	
O5	−0.6952	−0.7040		−0.6891	−0.7161	
O_a(-Li)	−0.6464	−0.5203	−0.5008	−0.7668	−0.6750	−0.6524
O_b(-Li)	−0.6464	−0.5485	−0.5290	−0.7668	−0.7292	−0.7067
C8	0.3758	0.3704		0.3840	0.3732	
C9	−0.2984	−0.2932		−0.2852	−0.2848	
C10	−0.2292	−0.2221		−0.1977	−0.2060	
C11	−0.3032	−0.2946		−0.2811	−0.2730	
C12	−0.1888	−0.1850		−0.1768	−0.1752	
C13	−0.2394	−0.2438		−0.2558	−0.2443	
$\sum C_{\text{Phenyl ring}}$	−0.8832	-0.8683		−0.8126	−0.8101	
C14		0.2622	0.2561		0.3691	0.2868
C15		0.3811	0.3900		0.3857	0.3453
C16		0.4102	0.4168		0.3945	0.4086
C17		0.3813	0.3900		0.3293	0.3897
C18		0.2628	0.2561		0.2977	0.3612
$\sum C_{\text{Croconic ring}}$		1.6976	1.709		1.7763	1.7916

increase of the number of C_5O_5, in the order LBSB, LCSB, LBCB, the main Li–O bond length also increases from 1.873 Å, 1.888 Å to 1.903 Å. As a result, the Li–O bond length gradually becomes larger and larger than the sum of ionic radii of the Li^+ and O atom. This clearly suggests the decreasing interactions between Li^+ and O atom.

3.2. Charges

Natural charge population (NPA) of LBSB, LCSB, LBCB and their corresponding free anions are listed in Table 2 to analyze the redistribution of electronic charge in the anion perturbed by the Li^+. The most striking feature observed is that oxygen atoms directly bounded with Li^+ have greater negative charge than that in the free anion (increase by 0.10–0.15e for NPA), while the metal-unbounded O atoms show less charge compared with the free anion. An interesting trend can be easily disclosed for these lithium salts, the positive charge on Li^+ increases from LBSB (0.9372), LCSB (0.9502) to LBCB (0.9525). This trend indicates very likely the decreasing electron transfer from anion (BSB^-, CSB^-, BCB^-) to Li^+. According to the study of Reed et al. [29], a change in population of 0.001e corresponds to the change of 0.001 hartree or 2.621 kJ/mol in energy stabilization. Then the largest electron transfer occurring in LBSB by 0.063e (NPA charge) corresponds to the largest charge-transfer interaction (E_{CT}) of 165.1 kJ/mol. This kind of charge-transfer interaction (E_{CT}) would be useful to study anion–cation binding interactions.

With the increase of the number of C_5O_5, in the order BSB^-, CSB^-, and BCB^-, the sum of charge population on $\sum C$ of the phenyl rings and $\sum C$ of the croconic rings change positively. The population on the phenyl and the croconic rings in their corresponding lithium salts has the same trend as the anions.

3.3. Energetics

The total energies and cation–anion binding energies of the lithium salts are listed in Table 3.

Several important trends concerning binding energies will be pointed out. First, with the increase of the number of C_5O_5, the calculated order of total and binding energies from B3LYP/6-311++G(3df,3pd)//B3LYP/6-31+G(d,p) calcu-

Table 3
Total, binding energies of the free anions, their lithium ion pairs at B3LYP/6-311++G(3df,3pd)//B3LYP/6-31+G(d,p) level[a]

	Total energy (a.u.)	Binding energies (KJ/mol)	Conductivity (298.2k, mS/cm)				
			PC	PC + DME	EC + DME	PC + THF	EC + THF
LBCB	−1166.227405	486	1.317	1.369	1.411	1.292	1.224
BCB^{-1}	−1158.757146						
LSCB	−1094.458195	538	1.088	1.275	1.335	1.224	1.182
SCB^{-1}	−1086.968135						
LBSB	−1022.680712	594	0.858	0.952	0.977	0.926	0.867
BSB^{-1}	−1015.169566						

[a] E_{Li+}, −7.284809 (a.u.), total energy of lithium cation is calculated at the same level.

Table 4
NBO energetic analysis for LBCB, LCSB, and LBSB at B3LYP/6-31+G(d,P) Level[a] (energies in kJ/mol)

	LBCB	LSCB	LBSB
$n^1o_1 \rightarrow n^*_M$	17.01	19.48	27.50
$n^2o_1 \rightarrow n^*_M$	15.38	17.05	5.31
$n^3o_1 \rightarrow n^*_M$	2.80	3.22	6.35
$n^1o_2 \rightarrow n^*_M$	18.56	18.14	27.50
$n^2o_2 \rightarrow n^*_M$	16.05	16.39	5.48
$n^3o_2 \rightarrow n^*_M$	2.97	3.01	6.27
E_{int}	72.77	77.29	78.41

[a] n^1O and n^2O refer to the p lone pairs of the oxygen atoms; n^*_M refers to the antibonding orbital of lone pair of the lithium cation.

lation is: LBSB > LCSB > LBCB. This order is in good agreement with general chemistry knowledge and demonstrates the weak coordinating strength of BCB anions.

Natural bond orbital (NBO) analysis at B3LYP/6-31+G(d,p) level is carried out to further our understanding of orbital interactions and charge delocalization. Of particular interests are the interactions between O (bounded with lithium cation) lone pairs and antibonding orbitals of lithium cation lone pair. The magnitudes of the interactions are shown in Table 4. It is interesting to notice that the calculated order of main orbital interaction energies (E_{int}) is the same as that of E_{bind} (Table 3), although E_{int} only accounts for part of the E_{bind}. In fact, the E_{int} could be roughly attributed to the covalency contributions to binding. Based on the charge delocalizing result, it is concluded that large E_{int} means strong charge delocalization from O atoms to cation, as well as less positive charges on cation (Table 4).

3.4. Properties

The motive driving us to initiate this investigation is to explore quantitatively the excellent chemical and electrochemical properties of LBCB, LCSB and LBSB, and to make some reasonable predictions according to the rules obtained through calculations. In this section, we will discuss the theoretical results and their comparisons with experimental data about their thermal stability, solubility, conductivity, electrochemical stability.

3.4.1. Thermal stability

There is now a general agreement to link the thermodynamic stability of a chemical species to the HOMO–LUMO gap increase [30]. These energy levels allow one to calculate a chemical reactivity index, the hardness (η) of these anions. Indeed, by applying Koopmans' theorem, Pearson showed that 2η is equal to the difference between the ionization potential (I) and electron affinity (A). The natural way to approximate the hardness in DFT is to evaluate it directly from the calculated ionization potential and electron affinity as [31].

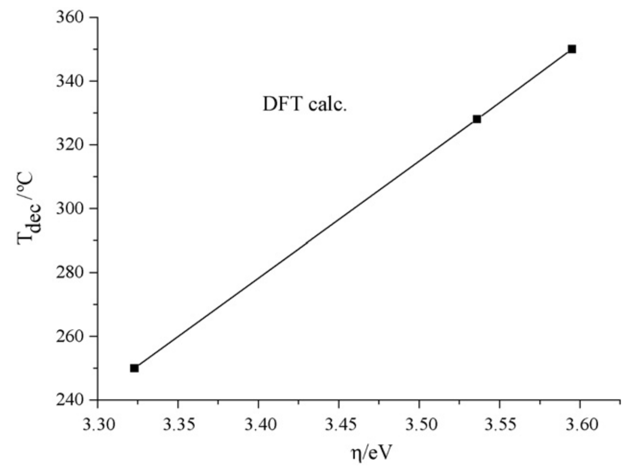

Fig. 2. Relations between the decomposition temperature (T_{dec}) of lithium salts and η of anions.

$$\eta = \frac{(I - A)}{2}$$

where I and A are obtained from total electronic energy calculations on the $N-1$, N, and $N+1$ electron systems at the neutral geometry.

$$I = E_{N-1} - E_N$$

$$A = E_N - E_{N+1}$$

Thus, the structures of BSB^-, CSB^-, and BCB^- were optimized by B3LYP/6-31+G(d,p), and the energy calculations were further performed by B3LYP/6-311++G(3df,3pd). The results are listed in Table 5. From these values, BSB^- appears the hardest base. According to the HSAB principle, the hard acid Li^+ prefers to coordinate with hardest base BSB^-, meaning that the most thermo stable lithium salt in the three salts is Li^+LBSB^-. We also notice (Fig. 2) that the η increases in parallel with the increase of decomposition temperature.

Table 5
Decomposing temperatures, energetics of anions by B3LYP at 6-311++G(3df,3pd) level

	E (a.u.)	I (a.u.)	A (a.u.)	η (eV)	T (°C) (in air)
BCB^0	−1158.547272				
BCB^{-1}	−1158.757146	0.209874	−0.034348	3.323	250
BCB^{-2}	−1158.722798				
SCB^0	−1086.776518				
SCB^{-1}	−1086.968135	0.191617	−0.068307	3.536	328
SCB^{-2}	−1086.899828				
BSB^0	−1014.994588				
BSB^{-1}	−1015.169566	0.174978	−0.089221	3.595	350
BSB^{-2}	−1015.080345				

3.4.2. Solubility

We have chosen to calculate a new, theoretical set of parameters and correlated them with the solubility of a lithium salt. While these parameters are derived solely from computational methods, we have deliberately chosen them to have an obvious correspondence with the parameters of linear solvation energy relationship (LSER). The most general theoretical linear solvation energy relationships (TLSER), developed by Cramer et al. [32], is thus:

$$\mathrm{Log}\,(S) = C_0 + C_1 V_{\mathrm{MC}} + C_2\pi^* + C_3\varepsilon_\alpha + C_4\varepsilon_\beta + C_5 q^+ + C_6 q^-$$

The TLSER volume term, V_{mc} is the molecular van der Waals volume calculated with the method of Hopfinger. The polarizability term, π^*, is derived from the polarization volume computed with the Kurtz's method. Just as with LSER, hydrogen bonding is divided into donor and acceptor components. Since all intermolecular interactions can be considered to have varying degrees of covalent and electrostatic components, separate descriptors have been chosen to address each of these two limiting formalisms. The covalent contribution to Lewis basicity, ε_β, is taken as the difference in energy between the lowest unoccupied molecular orbital (E_{LUMO}) of the solvent propylene carbonate (PC) and the highest occupied molecular orbital (E_{HOMO}) of the solute. Thus, smaller values of ε_β denote a greater covalent basicity. The electrostatic basicity contribution is denoted by q^-, the magnitude of the most negative atomic partial charge in the molecule. Analogously, the hydrogen bonding acidity is divided into two components: ε_α is the energy difference between E_{HOMO} of PC and E_{LUMO} of the solute and q^+ is the magnitude of the most positively charged hydrogen atom in the molecule.

The equation has been applied to the three anions (BCB^-, SCB^-, and BSB^-). With B3LYP/6-311++G(3df,3pd)//B3LYP/6-31+G(d,p)-derived descriptors (Table 6), we obtain a result that the solubility S of a lithium salt to be quite sensitive to anion volume, polarizability, and hydrogen bonding acidity ε_α. The regression is of:

$$\mathrm{Log}\,(S) = 11.9741 - 0.0018 V_{\mathrm{MC}} - 0.0989\pi^* - 1.1752\varepsilon_\alpha - 0.6121\varepsilon_\beta$$

The equation is significant according to F measure, the coefficients of V_{MC}, π^*, and ε_α are all Significant according to t-score, while that of ε_β is not.

Table 6
Solubility, descriptors of anions at B3LYP/6-311++G(3df,3pd) level

	BCB^{-1}	SCB^{-1}	BSB^{-1}
S (m, PC, 298.2k)	0.10	0.13	1.0
Log (S)	−1	−0.88606	0
$V\,\acute{\mathrm{Å}}^{-3}$	489.55	523.33	612.77
π^*	192.32	203.18	210.50
$\varepsilon_{\alpha(\mathrm{eV})}$	−7.78061	−8.52757	−9.88162
$\varepsilon_{\beta(\mathrm{eV})}$	3.64826	3.04416	2.75654
q^-	−0.64977	−0.71961	−0.71742

3.4.3. Conductivity

The disassociation of a lithium salt in the electrolyte solution is one of the key elements affecting Li-ion battery performance. It determines the number of free ions in an electrolyte, and thus the electric conductivity. Ue and Mori analyzed the contribution of the number of ions and the ion mobility to the electric conductivity for a variety of Li-ion nonaqueous electrolytes [33]. They found that the contribution from the number of free ions is generally larger than that from the ion mobility. Thus, the number of ions available plays a major role in determining the electric conductivity of a cell, and hence in the overall battery performance.

As generally recognized, the weaker the coordinating, the easer the separating of an ion-pair. With the increase of the number of C_5O_5, the calculated order of binding energies is $E_{\mathrm{bind}}(\mathrm{LBSB}) > E_{\mathrm{bind}}(\mathrm{LSCB}) > E_{\mathrm{bind}}(\mathrm{LBCB})$ (Table 3). Meanwhile, the Li–O bond length gradually becomes longer with this sequence (Table 1). Thus, the electron transfer from anion (BSB^-, SCB^-, BCB^-) to Li^+ becomes lesser according to NPA (Table 2) and the effective cation charge becomes larger (Table 2). Therefore, the BCB^- is the weakest anion of the three salts and Li^+LBCB^- would be disassociated more than the other two salts at the same concentration of electrolyte solutions. Compared with the calculated data, the experimental conductivity (Table 3) increases with the Li–O bond length and the effective cation charge, and decreases with E_{bind} and the extent of electron transfer from anion to cation.

3.4.4. Electrochemical stability window

Electrochemical reactions can be achieved by a direct electron transfer at a certain potential. If electrons are transferred from the HOMO of a compound, a correlation is expected between the E_{HOMO} and the oxidation potential E_{ox}, because the negatives of the orbital energies in the ground state are equal to the ionization potentials according to Koopman's theorem. Kita et al. correlated the limiting E_{ox} of five lithium tetraarylborates in 1,3-dioxolane/1,2-dimethoxyethane mixed solvents [34] with their respective HOMO energies calculated using a semi-empirical molecular orbital method (MNDO). Barthel et al. also proved this linear relationship using MNDO and AM1 for $B(O_2R)_2^{-1}$ (R = C_6H_4, C_6H_3F, C_6F_4, $C_{10}H_6$) [8].

3.4.4.1. Ab initio molecular orbital calculation. The limiting E_{ox} of the three anions in PC have been determined with the anodic stability of anions in the following order: $BSB^{-1} < SCB^{-1} < BCB^{-1}$ (Table 7). To examine this result, these HOMO energies were calculated using *ab initio* molecular orbital theory. The structure was optimized by HF/6-311++G(3df,3pd)//B3LYP/6-31+G(d,p). The results are given in Fig. 3, which shows a relatively good linear relationship between the HOMO energies and the limiting oxidation potentials. The regression result is:

$$E_{\mathrm{ox}} = -1.23734 - 1.01472 \times E_{\mathrm{HOMO}}$$

$$(R = 0.9959,\ \mathrm{S.D.} = 0.07877)$$

Table 7
Limiting oxidation potentials, ionization potential calculations using DFT methods at 6-311++G(3df,3pd) level and HOMO energies using HF methods at 6-311++G(3df,3pd) level

Anion	BCB^-	SCB^-	BSB^-
$E_{radical}$ (a.u.)	−1158.547272	−1086.776518	−1014.994588
E_{anion} (a.u.)	−1158.757146	−1086.968135	−1015.169566
ΔE (KJ/mol)	550.49743	502.60950	458.96556
$ZPE_{radical}$ (KJ/mol)	0.098055	0.147378	0.199818
ZPE_{anion} (KJ/mol)	0.101005	0.151004	0.201203
ΔZPE (KJ/mol)	7.73782	9.51096	3.63284
I_p (eV)	5.631	5.116	4.724
E_{HOMO} (eV)	−6.845	−6.044	−5.638
E_{OX} (V vs. Li^+/Li)	5.5	4.8	4.3

3.4.4.2. DFT calculations. To further examine this experimental result, the ionization potentials were also calculated by *ab initio* density functional theory. This method usually gives higher accuracy in energy calculations, where the relative accuracies of various different model chemistries were considered by their performances on the G2 molecule sets [35]. The structural optimization of the anion was carried out by B3LYP/6-31+G(d,p) followed by the energy calculations on higher basis sets, 6-311++G(3df,3pd). Because Koopman's theorem is not valid in the DFT scheme, the adiabatic ionization potential Ip was calculated from the energy difference ΔE between the total energy of the anion E_{anion} and that of the neutral radical $E_{radical}$ generated by one-electron oxidation, as shown in Fig. 4. The structure and energy of the neutral radical was calculated by the same methods. Zero-point energies (ZPE) were also calculated by frequency analysis using only B3LYP/6-31+G(d,p). The ionization potentials were calculated from the following equation:

$$I_p = \Delta E - \Delta ZPE$$

The results are summarized in Table 7 and Fig. 5. The ionization potentials obtained by DFT methods were about 0.9–1.2 eV lower than those predicted from Koopman's theorem by *ab initio* MO methods. This is because Koopman's theorem neglects electron relaxation and electron correlation, which is important for the current systems containing anions. Fig. 5 shows good correlation between I_p and E_{ox}, completely confirming the experimental result: $BSB^{-1} < SCB^{-1} < BCB^{-1}$. The regression result is:

$$E_{ox} = 1.4849 + 0.75454 \times I_p$$

$$(R = 0.99984,\ S.D. = 0.01145)$$

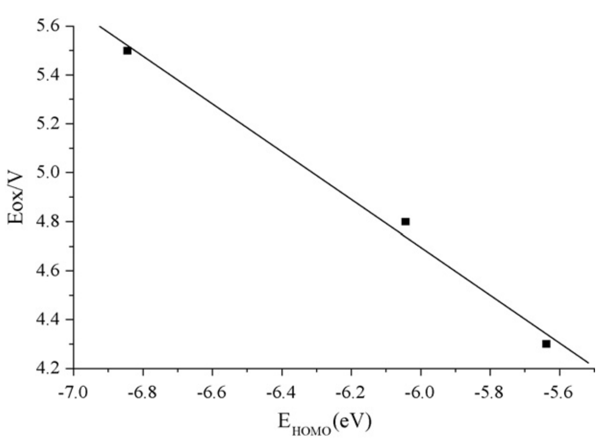

Fig. 3. Relation between E_{ox} and E_{HOMO} of anions at HF/6-311++G(3df,3pd) level.

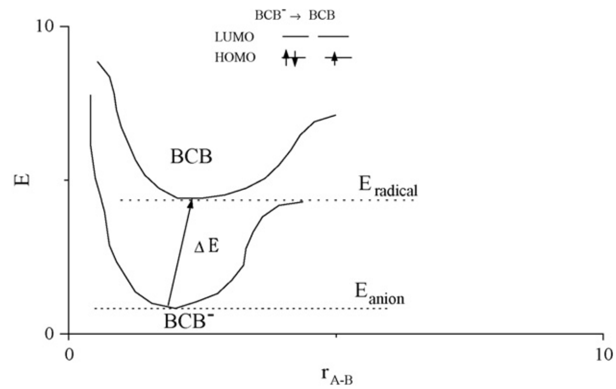

Fig. 4. Procedure to calculate the ionization potentials of anions.

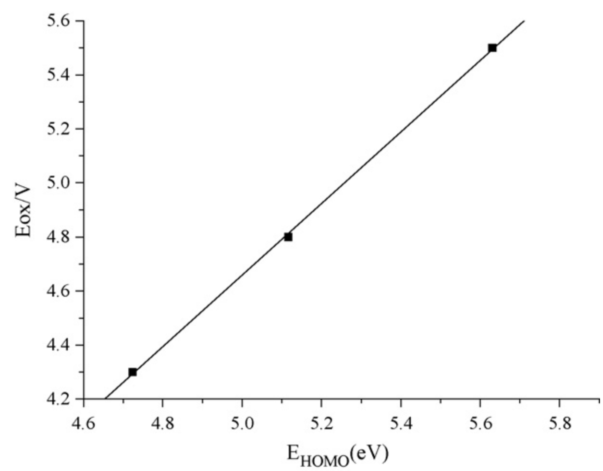

Fig. 5. Relation between E_{ox} and I_p of anions at DFT B3LYP/6-311++G (3df,3pd) level.

4. Conclusions

We have presented a DFT study of the structures and cation–anion interactions of Li^+BCB^- and its derivatives, Li^+LSCB^- and Li^+LBSB^-. Energetic calculations suggest that tetrahedron structures are the most stable. NBO analyses indicate strong delocalization interactions involving the lone pairs of metal-chelated oxygen atoms and the antibonding orbitals of cations. The LBSB holds the strongest charge delocalization from O atoms to lithium cation among these salts. Comparisons of binding energies and E_{int} energies among these lithium complexes show that LBSB hold stronger Lewis acidity than LSCB, or LBCB. The excellent correlations were observed between thermal stability and the energy difference between I and A, solubility and polarizability, ionic conductivity and binding energy, electrochemical stability and E_{HOMO}, or ionization potential (I_{p}).

Acknowledgments

This work was supported by Chinese Academy of Sciences Hundred Talents program and the National Science Foundation of China (Grant No. 50372064).

References

[1] N. Nanbu, K. Tsuchiya, T. Shibazaki, Y. Sasaki, Electrochem. Solid-State Lett. 5 (9) (2002) A202.
[2] M. Handa, M. Suzuki, J. Suzuki, H. Kanematsu, Y. Sasaki, Electrochem. Solid-State Lett. 2 (2) (1999) 60.
[3] X.M. Wang, E. Yasukawa, S. Kasuya, J. Electrochem. Soc. 147 (7) (2000) 2421.
[4] Wu. Xu, C. Austen Angell, Electrochem. Solid-State Lett. 4 (1) (2001) E1.
[5] J. Barthel, A. Schmid, H.J. Gores, J. Electrochem. Soc. 147 (1) (2000) 21.
[6] J. Barthel, R. Buestrich, H.J. Gores, M. Schmidt, M. Wühr, J. Electrochem. Soc. 144 (1997) 3866.
[7] J. Barthel, M. Wühr, R. Buestrich, H.J. Gores, J. Electrochem. Soc. 142 (8) (1995) 2527.
[8] J. Barthel, R. Buestrich, E. Carl, H.J. Gores, J. Electrochem. Soc. 143 (11) (1996) 3572.
[9] M. Handa, S. Fukuda, Y. Sasaki, K. Usami, J. Electrochem. Soc. 144 (9) (1997) L235.
[10] J. Barthel, R. Buestrich, H.J. Gores, M. Schmidt, M. Wu, J. Electrochem. Soc. 144 (1997) 3866.
[11] W. Xu, C.A. Angell, Electrochem. Solid-State Lett. 4 (2001) E1.
[12] M. Ue, A. Murakami, S. Nakamura, J. Electrochem. Soc. 14 (12) (2002) A1572.
[13] K. Tasaki, J. Electrochem. Soc. 149 (4) (2002) A418.
[14] P. Johansson, S.P. Gejji, J. Tegenfeldta, J. Lindgren, Electrochim. Acta l.43 (10–11) (1998) 1375.
[15] Z.-M. Xue, C.-H. Chen, Electrochim. Acta 49 (2004) 5167.
[16] Z.M. Xue, C.H. Chen, Prog. Chem. (Chinese) 17 (3) (2005) 399.
[17] Z.M. Xue, B. Liu, C.H. Chen, Electrochim. Acta 51 (2006) 4554.
[18] Z.M. Xue, J.J. Cheng, C.H. Chen, J. Mol. Struct.: THEOCHEM 763 (2006) 181.
[19] Z.M. Xue, C.H. Chen, Mol. Simul. 32 (2006) 401.
[20] Z.M. Xue, C.H. Chen, J. Power Sources 171 (2007) 944.
[21] C. Lee, W. Yang, R.G. Parr, Phys. Rev. B 37 (1988) 785.
[22] P.J. Hay, W.R. Wadt, J. Chem. Phys. 82 (1985) 270.
[23] E.D. Glendening, A.E. Reed, J.E. Carpenter, F. Weinhold. NBO Version 3.1.
[24] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, J.A. Montgomery Jr., T. Vreven, K.N. Kudin, J.C. Burant, J.M. Millam, S.S. Iyengar, J. Tomasi, V. Barone, B. Mennucci, M. Cossi, G. Scalmani, N. Rega, G.A. Petersson, H. Nakatsuji, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, M. Klene, X. Li, J.E. Knox, H.P. Hratchian, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J.W. Ochterski, P.Y. Ayala, K. Morokuma, G.A. Voth, P. Salvador, J.J. Dannenberg, V.G. Zakrzewski, S. Dapprich, A.D. Daniels, M.C. Strain, O. Farkas, D.K. Malick, A.D. Rabuck, K. Raghavachari, J.B. Foresman, J.V. Ortiz, Q. Cui, A.G. Baboul, S. Clifford, J. Cioslowski, B.B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R.L. Martin, D.J. Fox, T. Keith, M.A. Al-Laham, C.Y. Peng, A. Nanayakkara, M. Challacombe, P.M.W. Gill, B. Johnson, W. Chen, M.W. Wong, C. Gonzalez, J.A. Pople, Gaussian 03, Revision C. 02, Gaussian, Inc., Wallingford CT, 2004.
[25] R. Arnaud, D. Benrabah, J.-Y. Sanchez, J. Phys. Chem. 100 (1996) 10882.
[26] J.S. Francisco, I.H. Williams, J. Phys. Chem. 91 (1990) 8522.
[27] M. Spoliti, N. Sanna, V.J. Di Martino, Mol. Struct. 258 (1992) 83.
[28] F. Ramondo, L. Bencivenni, R. Caminiti, C. Sadun, Chem. Phys. 151 (1991) 179.
[29] A.E. Reed, L.A. Curtiss, F. Weinhold, Chem. Rev. 88 (1988) 899.
[30] R.G. Pearson, Proc. Natl. Acad. Sci. U.S.A. 89 (1996) 1827.
[31] D.J. Tozer, F. Proft De, J. Phys. Chem. A 109 (2005) 8923.
[32] C.J. Cramer, G.R. Famini, A.H. Lowrey, Acc. Chem. Res. 26 (1993) 599.
[33] M. Ue, S. Mori, J. Electrochem. Soc. 141 (1994) 3336.
[34] F. Kita, A. Kawakami, T. Sonoda, H. Kobayashi, New Sealed Rechargeable Batteries and Supercapacitors, PV93-23, in: B.M. Barnett, E. Dowgiallo, G. Halpert, Y. Matsuda, Z.-i. Takehara (Eds.), The Electrochemical Society Proceedings Series, Pennington, NJ, 1993, p. 321.
[35] J.B. Foresman, A. Frisch, Exploring Chemistry with Electronic Structure Methods, second ed., Gaussian, Pittsburgh, PA, 1996.